Spatial Statistics
- Research

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1 A General Periodic Fourier–State-Space Spatio–Temporal Model

1.1 Model setup

We consider a real-valued Gaussian spatio–temporal process

$$\{X_t(x) : x \in [0, 2\pi],\ t \in \mathbb{Z}\},$$

defined on a one-dimensional spatial domain with periodic boundary conditions (a circle) and discrete time. For each time t, the spatial field $X_t(\cdot)$ is represented by a truncated Fourier series

$$X_t(x) \ = \ a_0(t) \ + \ \sum_{j=1}^{J} \Big[a_j(t) \cos(jx) + b_j(t) \sin(jx) \Big], \qquad x \in [0, 2\pi], \tag{1}$$

where J is a truncation level and

$$a_j(t),\ b_j(t), \quad j = 0, \dots, J,\ t \in \mathbb{Z},$$

are jointly Gaussian random coefficients.

For $j \geq 1$, the pair $(a_j(t), b_j(t))$ corresponds to the jth spatial frequency (wavenumber). Low j terms encode large-scale, slowly varying spatial structure, while high j terms encode fine-scale spatial detail.

1.2 Spatial margin: Matérn-like variances

The spatial covariance at each fixed time is controlled by a Matérn-like spectral density in the Fourier domain. For $j = 0, \dots, J$ we specify

$$\sigma_j^2 \ \propto \ \left(\kappa^2 + j^2\right)^{-(\nu+1/2)}, \quad \kappa = \frac{\sqrt{2\nu}}{\ell}, \tag{2}$$

where

- $\nu > 0$ is the Matérn smoothness parameter,
- $\ell > 0$ is a spatial range parameter,
- $\kappa = \sqrt{2\nu}/\ell$ is an inverse scale parameter.

NOTE: This spectral form is appropriate for a *non-periodic* domain $\mathbb{R}$. On a periodic domain $[0, 2\pi]$, the correspondence between this discrete spectrum and the continuous Matérn model requires careful treatment.

Question: Should we (a) sample the continuous Matérn spectrum at integer frequencies j, or (b) explicitly periodize the Matérn covariance function and compute its Fourier coefficients?

For simulation we normalize these variances so that the total marginal variance of $X_t(x)$ is σ^2_{tot}. Defining

$$\tilde{\sigma}_j^2 \;=\; \left(\kappa^2 + j^2\right)^{-(\nu+1/2)},$$

we set

$$\sigma_j^2 \;=\; \sigma^2_{\text{tot}} \cdot \frac{\tilde{\sigma}_j^2}{\sum_{k=0}^{J} \tilde{\sigma}_k^2}, \qquad j = 0, \dots, J. \tag{3}$$

1.3 Temporal dynamics: block-diagonal AR(1) with rotation and correlated innovations

We collect the Fourier coefficients into a state vector

$$\mathbf{Z}_t \;=\; \big(a_0(t), a_1(t), \dots, a_J(t), b_1(t), \dots, b_J(t)\big)^\top \;\in\; \mathbb{R}^{2J+1}. \tag{4}$$

The temporal evolution is governed by a *block-diagonal* linear Gaussian state-space model. For the constant mode $j = 0$, we have a scalar AR(1):

$$a_0(t+1) = \rho_0\, a_0(t) + \eta_{0,t}, \qquad \eta_{0,t} \sim \mathcal{N}\big(0, \sigma_0^2(1 - \rho_0^2)\big), \tag{5}$$

with $|\rho_0| < 1$ ensuring stationarity.

For each mode $j \geq 1$, the cosine and sine coefficients evolve jointly according to a 2×2 block AR(1):

$$\begin{pmatrix} a_j(t+1) \\ b_j(t+1) \end{pmatrix} = M_j \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} + \begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix}, \tag{6}$$

where the 2×2 transition matrix is

$$M_j = \rho_j \begin{pmatrix} \cos\theta_j & -\sin\theta_j \\ \sin\theta_j & \cos\theta_j \end{pmatrix}, \tag{7}$$

with $\rho_j \in (0, 1)$ (amplitude persistence) and $\theta_j \in \mathbb{R}$ (rotation angle).

The 2×2 innovation covariance for mode j is

$$\begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix} \sim \mathcal{N}\Big(\mathbf{0},\; \Sigma_{\varepsilon,j}\Big), \qquad \Sigma_{\varepsilon,j} = \sigma_j^2(1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}, \tag{8}$$

where $\phi_j \in (-1, 1)$ is the *within-frequency correlation parameter* between the cosine and sine innovations.

WARNING — EMPIRICAL VALIDATION ISSUE: The Lyapunov equation $\Sigma_j = M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j}$ with target stationary covariance $\Sigma_j = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}$ does yield (8) as shown in Section 1.4. However, empirical validation showed:

- Empirical $\mathrm{Cor}(a_1, b_1) = 0.208$, theoretical $\phi_1 = 0.696$ (70% error)
- Phase drift empirically ≈ 0 even when $\theta_j \neq 0$ was set
- ACF of amplitude decays faster than ρ_j^k (up to 57% error for high j)

This suggests either (a) an implementation error in the R code, or (b) a misunderstanding of what quantity should match theory when rotation is present.

The innovations are independent across different frequencies j and across time t.

Physical interpretation

Rotation parameter θ_j: When $\theta_j = 0$, the transition matrix $M_j = \rho_j I_2$, so cosine and sine evolve independently. When $\theta_j \neq 0$, we have temporal mixing:

$$a_j(t+1) = \rho_j[\cos\theta_j \cdot a_j(t) - \sin\theta_j \cdot b_j(t)] + \eta_{a,j,t},$$

inducing *phase rotation*: spatial patterns propagate around the circle over time. The eigenvalues of M_j are $\rho_j e^{\pm i\theta_j}$, so stationarity holds for $|\rho_j| < 1$ regardless of θ_j.

Question: Does phase rotation θ_j have a physical interpretation in terms of wave propagation speed on the circle? If $\theta_j = c \cdot j$ for constant c, does this correspond to a traveling wave with speed c?

Innovation correlation parameter ϕ_j: When $\phi_j = 0$, the innovations $\eta_{a,j,t}$ and $\eta_{b,j,t}$ are independent. When $\phi_j \neq 0$, the innovations are correlated: $\mathrm{Cov}(\eta_{a,j,t}, \eta_{b,j,t}) = \sigma_j^2(1-\rho_j^2)\phi_j$. At stationarity, this targets $\mathrm{Cov}(a_j(t), b_j(t)) = \sigma_j^2\phi_j$ via the Lyapunov equation (Section 1.4).

Typical parameterization: $\phi_j = \phi_0/(1+j)$ or $\phi_j = \phi_0 e^{-\gamma j}$.

Combined effect: $\theta_j \neq 0$ creates *temporal cross-dependence*; $\phi_j \neq 0$ creates *contemporaneous correlation*. Together they allow rich within-frequency dynamics while maintaining the block-diagonal structure.

Key empirical finding: Simulated $\mathrm{Cor}(a_j, b_j)$ showed 70% error for $j = 1$ and wild scatter for other modes, suggesting either incorrect initial condition draws or a numerical issue in the multivariate normal generation.

1.4 Stationary covariance of $(a_j(t), b_j(t))$ via Lyapunov equation

At stationarity, the 2×2 covariance matrix

$$\Sigma_j = \mathrm{Cov}\begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix}$$

satisfies the discrete-time Lyapunov equation

$$\Sigma_j = M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j}. \tag{9}$$

Corrected derivation: For the rotation-scaling matrix (7), we have

$$M_j M_j^\top = \rho_j^2 I_2,$$

so (9) becomes

$$\Sigma_j = \rho_j^2 \Sigma_j + \Sigma_{\varepsilon,j} \quad \Longrightarrow \quad \Sigma_{\varepsilon,j} = (1 - \rho_j^2)\Sigma_j.$$

If we target the stationary covariance

$$\Sigma_j = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix},$$

then the innovation covariance must be

$$\Sigma_{\varepsilon,j} = \sigma_j^2 (1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}.$$

This formula is *independent of* θ_j because $M_j M_j^\top = \rho_j^2 I_2$ for any rotation angle. Therefore equation (8) is theoretically correct, and the empirical mismatch must arise from implementation or interpretation errors.

Advisor questions:

1. The empirical ACF of amplitude $R_j(t) = \sqrt{a_j(t)^2 + b_j(t)^2}$ decays much faster than ρ_j^k (up to 57% error for high j). Should the theoretical ACF be computed for the *amplitude* instead of the coefficients?

2. The empirical correlation $\mathrm{Cor}(a_j(t), b_j(t))$ showed wild scatter instead of matching $\phi_j \approx 0.65$–0.70. Is this a finite-sample effect, or is there a bias in how the stationary initial condition is drawn?

3. The spatial covariance at large lags showed an "upturn" (U-shape) instead of monotonic decay. Is this the "ridge" artifact from nonseparable models warned about in Stein (2005)? Does this indicate we need different temporal structure?

Therefore, at stationarity:

$$\mathrm{Var}(a_j(t)) = \sigma_j^2, \tag{10}$$

$$\mathrm{Var}(b_j(t)) = \sigma_j^2, \tag{11}$$

$$\mathrm{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j. \tag{12}$$

The stationary covariance depends on ϕ_j but *not* on θ_j. The rotation angle affects temporal evolution but not marginal variances or contemporaneous correlation.

1.5 Nonseparability via frequency-dependent persistence

The key nonseparable feature arises from allowing ρ_j to depend on spatial frequency j. We propose an *algebraic (power-law) decay* parameterization

$$\rho_j = \frac{1}{(1+\lambda j^{\alpha})^{\beta}}, \qquad j = 1, \dots, J, \tag{13}$$

with $\rho_0 \in (0,1)$ set separately, and parameters $\lambda > 0$, $\alpha > 0$, $\beta > 0$.

Under this structure, low-frequency modes have $\rho_j \approx 1$ (slow temporal decay), while high-frequency modes have smaller ρ_j (fast decay). The effective spatial correlation structure changes with time lag: at short lags both low and high frequencies contribute, but at long lags only low frequencies remain. This space–time covariance cannot be written as $C_{\text{space}}(h)C_{\text{time}}(k)$ and is therefore nonseparable.

1.6 Space–time covariance

Under the block-diagonal structure, the space–time covariance

$$C(h,k) = \mathbb{E}\big[X_t(x)X_{t+k}(x+h)\big]$$

can be computed when different modes are uncorrelated across frequencies. For $\theta_j = 0$ (no rotation), we have

$$\mathbb{E}[a_j(t)a_j(t+k)] = \mathbb{E}[b_j(t)b_j(t+k)] = \sigma_j^2\rho_j^{|k|},$$

yielding

$$C(h,k) = \sum_{j=0}^{J} \sigma_j^2\rho_j^{|k|}\cos(jh). \tag{14}$$

Derivation assumes $\theta_j = 0$: When M_j includes rotation, the lag-k covariance involves M_j^k:

$$M_j^k = \rho_j^k \begin{pmatrix} \cos(k\theta_j) & -\sin(k\theta_j) \\ \sin(k\theta_j) & \cos(k\theta_j) \end{pmatrix},$$

so

$$\mathbb{E}[a_j(t)a_j(t+k)] = \sigma_j^2\rho_j^k\cos(k\theta_j) + \text{cross-terms involving } \phi_j.$$

The spatial covariance (14) is therefore correct *only when* $\theta_j = 0$ or when cross-terms cancel in the Fourier reconstruction. **please verify when rotation is present.**

1.7 Parameter overview

The full model is specified by the following parameters, organized by their role in the model structure.

Fourier truncation

- $J \in \mathbb{Z}_+$: highest Fourier mode (truncation level). Controls the finest spatial scale represented. Larger J allows finer spatial detail; in practice J is chosen so that the total variance in modes $j > J$ is negligible.

Spatial margin parameters (Matérn-like spectrum)

- $\nu > 0$: Matérn smoothness parameter. Larger ν gives smoother spatial fields. Sample paths are $\lceil \nu \rceil - 1$ times mean-square differentiable.
- $\ell > 0$: spatial range parameter. Larger ℓ increases the spatial correlation length by shifting spectral mass to lower frequencies.
- $\sigma^2_{\text{tot}} > 0$: total marginal variance of the field $X_t(x)$. Overall scale parameter.
- $\kappa = \sqrt{2\nu}/\ell$: (derived) inverse range parameter used in the spectral density (2).

The mode variances σ_j^2 for $j = 0, \ldots, J$ are derived from $(\nu, \ell, \sigma^2_{\text{tot}})$ via (2) and (3).

Temporal persistence parameters

- $\rho_0 \in (0, 1)$: AR(1) persistence for the constant mode $a_0(t)$. Set separately to avoid nonstationarity (the formula (13) would give $\rho_0 = 1$ for $j = 0$).
- $\lambda > 0$: controls overall temporal decay strength across frequencies in (13).
- $\alpha > 0$: controls how strongly persistence changes with frequency j in (13).
- $\beta > 0$: decay exponent in (13). Smaller β gives slower (longer-memory) decay.

The mode-specific persistence parameters ρ_j for $j = 1, \ldots, J$ are derived from (λ, α, β) via (13).

Phase rotation parameters

- $\theta_j \in \mathbb{R}$ for $j = 1, \ldots, J$: rotation angle for mode j in the 2×2 transition matrix M_j (equation (7)).
- When $\theta_j = 0$ for all j, the model reduces to independent cosine/sine evolution (no temporal mixing).
- Typical parameterization: $\theta_j = \theta_0/j$ or $\theta_j = \theta_0/(1 + j)$ for a scalar hyperparameter $\theta_0 \in \mathbb{R}$, so that lower frequencies rotate more than higher frequencies.

Empirical finding: Phase drift validation showed mean $\Delta\text{phase} \approx 0$ even when θ_j was set to nonzero values. This suggests either (a) rotation is not being applied correctly in simulation, or (b) the phase of amplitude $R_j(t) = \sqrt{a_j^2 + b_j^2}$ does not drift at rate θ_j as expected. **What is the theoretical expected phase drift rate when rotation is present?**

Innovation correlation parameters

- $\phi_j \in (-1, 1)$ for $j = 1, \ldots, J$: within-frequency correlation between innovations $\eta_{a,j,t}$ and $\eta_{b,j,t}$ (equation (8)).
- When $\phi_j = 0$ for all j, the innovations are independent (cosine and sine uncorrelated at the same time).
- When $\phi_j \neq 0$, the model induces contemporaneous correlation $\mathrm{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j$ at stationarity.
- Typical parameterization: $\phi_j = \phi_0/(1 + j)$ or constant $\phi_j = \phi_0$ for all j, where $\phi_0 \in (-1, 1)$ is a scalar hyperparameter.

Empirical finding: Simulated $\mathrm{Cor}(a_j, b_j)$ showed 70% error for $j = 1$ (empirical 0.208 vs. theoretical 0.696) and high scatter for other modes. This suggests either (a) the stationary initial condition is drawn incorrectly, or (b) there is a numerical issue in the multivariate normal generation. The Lyapunov theory is correct (Section 1.4), so the issue is implementation-related.

Time horizon

- $T \in \mathbb{Z}_+$: number of time points to simulate or observe. Not a model parameter but a data specification.

Parameter count summary

For a model with Fourier truncation J and the simplest scalar hyperparameter choices, the total number of free parameters is:

- **Minimal model** (independent modes, no rotation, no innovation correlation): $J = 50$ modes

$$\text{Parameters: } \nu, \ell, \sigma^2_{\text{tot}}, \rho_0, \lambda, \alpha, \beta \quad \Rightarrow \quad 7 \text{ parameters total.}$$

- **With rotation** (add θ_0 for $\theta_j = \theta_0/j$):

$$\text{Parameters: } 7 + 1 = 8.$$

- **With rotation and innovation correlation** (add θ_0 and ϕ_0 for $\phi_j = \phi_0/(1 + j)$):

$$\text{Parameters: } 7 + 1 + 1 = 9.$$

- **Fully flexible** (separate θ_j and ϕ_j for each mode $j = 1, \ldots, J$):

$$\text{Parameters: } 7 + J + J = 7 + 2J.$$

For $J = 50$, this is 107 parameters—likely overparameterized for most applications.

Question: For inference (e.g., Whittle likelihood in Fourier space), which parameterization strikes the best balance between flexibility and identifiability? Should we use mode-specific $\{\rho_j, \theta_j, \phi_j\}$ as free parameters, or constrain them via smooth functions of j with scalar hyperparameters?

By setting $\theta_j = 0$ and $\phi_j = 0$ for all j, the model reduces to the simple independent-mode AR(1) with uncorrelated cosine and sine components (7 parameters). Nonzero θ_j and ϕ_j introduce richer within-frequency dynamics: temporal mixing and phase rotation (θ_j), and instantaneous correlation (ϕ_j).

Open Questions

1. **Periodic Matérn spectrum:** Should we sample the continuous Matérn spectral density at integer frequencies j, or explicitly periodize the Matérn covariance and compute its Fourier series?

2. **Rotation and ACF:** When M_j includes rotation ($\theta_j \neq 0$), what is the theoretical ACF of amplitude $R_j(t) = \sqrt{a_j^2 + b_j^2}$? Does it match ρ_j^k, or does rotation change the decay rate?

3. **Innovation correlation identifiability:** The empirical $\mathrm{Cor}(a_j, b_j)$ showed high variance. Is ϕ_j identifiable from data, or does it trade off with θ_j?

4. **Phase drift interpretation:** If $\theta_j = c \cdot j$, does this correspond to a traveling wave with speed c? Physical meaning of frequency-dependent rotation?

5. **Cross-frequency dependence:** Current model has block-diagonal dynamics. Should we add off-block-diagonal elements in M or Σ_ε for cross-frequency coupling? What structure preserves computational efficiency?

6. **Parameter parsimony for inference:** For Whittle likelihood estimation, should we use mode-specific $\{\rho_j, \theta_j, \phi_j\}$ as free parameters, or constrain via smooth functions with scalar hyperparameters (e.g., $\theta_j = \theta_0/j$)? What is the trade-off between flexibility and identifiability?

2 Next steps

Write the Covariance in closed form as then it nice to simulate with fft due to circulant structure.

If we truncate to J it depends on how dense the items are

Want $Z(x) = \sum a_j(t)\cos(jx) + b_j(t)\sin(jx)$ to be a Gaussian process with covariance $C(h, k)$.

Consider Aliasing $Z(k/n) = \sum a_j(t)\cos(2\pi jk/n) + b_j(t)\sin(2\pi jk/n)$

Now simulat $A'_j = \sum_p A_{a_j+pn}$ and $B'_j = \sum_p B_{b_j+pn}$

and $\mathrm{Var}(Z(k/n)) = \sum_{p=-\infty}^{\infty} \sigma^2_{j+pn} = \sum_{p=-q}^{q} \text{exact} + \sum_{p=q+1}^{\infty} + \sum_{p=-\infty}$ tail by approximation

Ask about the proof for the other of psd is $\int_d$